

Rapport-preuve - NS Methods (provenance : évalué vs mentionné)

Corpus : **55 articles**. Évalués validés vs tables_only, mentionnés vs no_tables (prose). Matching déterministe.

Scores

Score	Valeur
Précision évalués	93.1%
Précision mentionnés	91.8%
Recall évalués (indicatif)	81.4%
Recall mentionnés (indicatif)	80.4%
Évalués vérifiés / mentionnés vérifiés	202 / 134

A. Preuve - modèles/méthodes ÉVALUÉS (202)

Article	Entité	Trouv é	Extrait (tableau)		
AdaptativeNS.md	Adaptive Negative Sampling	oui	00016 # Knowledge Graph Embedding via Adaptive Negative Subsampling Dong Zhu ¹ *, Haona		
ADNS.md	Affinity Dependent Negative Sampling	oui	CEUR-WS.org/Vol-2635/paper8.pdf # Affinity Dependent Negative Sampling for Knowledge Graph E		
ADNS.md	Distributional Negative Sampling	oui	nent candidate as replaceable entity is Distributional Negative Sampling [9]. This approach do		
ADNS.md	Random Negative Sampling	oui	with distributed negative sampling and random negative sampling and the results have		
ASA.md	Adaptive Self-Adversarial Negative Sampling	oui	lation-aware Graph Attention Model With Adaptive Self-adversarial Training *Xiao Qin ¹ , Nasrullah Shei		
ASA.md	ASA■■■■	oui	technique - adaptive self-adversarial (ASA) negative sampling. ASA reduces the fal		
ASA.md	NSCaching	oui	-adversarial negative sampling method NSCaching as the baseline. It provides some fle		
ASA.md	Random Negative Sampling	oui	for link/relationship prediction task. Random negative sampling has been widely adopt		
BatchNS.md	Batched Negative Sampling	oui	D. ⁵In fact, our approach to batched negative sampling, described in Section		
CAKE.md	CAKE	oui	Xiv:2202.13785v3 [cs.AI] 17 Apr 2022 # CAKE: A Scalable Commonsense-Aware Framework		
CAKE.md	Commonsense-Aware Negative Sampling	oui	[cs.AI] 17 Apr 2022 # CAKE: A Scalable Commonsense-Aware Framework For Multi-View Knowledge Grap		

Article	Entité	Trouv é	Extrait (tableau)		
CAKE.md	Domain-based Negative Sampling	oui	oring function without a generator. (3) Domain-based sampling: domain-based NS (Wang e		
CAKE.md	MVLP	oui	ule and the multi-view link prediction (MVLP) module. Firstly, the ACG module extrac		
CAKE.md	None Sampling	oui	improve the efficiency of sampling. (5) None-sampling: NS-KGE (Li et al., 2021) elim		
CAKE.md	NSCaching	oui	correct domain. (4) Efficient sampling: NSCaching (Zhang et al., 2019b) employs cache con		
CAKE.md	Self-Adversarial Negative Sampling	oui	s in an adversarial training framework. Self-adversarial sampling (Sun et al., 2019) performs si		
CAKE.md	Uniform Negative Sampling	oui	or relation in a positive triple from a uniform distribution (Wang et al., 2014). (2) A		
CANS.md	Confidence-Aware Negative Sampling	oui	national Conference on Big Knowledge # Confidence-Aware Negative Sampling Method for Noisy Know		
CCS.md	Bernoulli Negative Sampling	oui	odel training. The TansH model uses the Bernoulli sampling method, which aims at reducing		
CCS.md	Cluster-Cache Sampling	oui	igh-quality negative triplets, based on cluster-cache sampling (CCS). It is a general negativ		
CCS.md	Kbgan Sampling	oui	arial training framework. IGAN [40] and Kbgan [41] both introduce GAN for negative sa		
CCS.md	NSCaching Sampling	oui	d increases the difficulty of training. NSCaching [38] has fewer parameters than both IGA		
ConceptDriven.md	Commonsense-Aware Negative Sampling	oui	^*\$. Similarly, [2] proposes a scalable commonsense-aware NS framework, to extract commonsense fr		
ConceptDriven.md	Concept Domain Negative Sampling	oui	ctive constraints. E.g., [4] proposes a concept domain negative sampling (CDNS) strategy, whic		
ConceptDriven.md	Concept-Driven Negative Sampling	oui	10.1109/DSC59305.2023.00047 # A Novel Concept-Driven Negative Sampling Mechanism for Enhanci		
ConceptDriven.md	Uniform Negative Sampling	oui	hten the best configuration. Under the "unif." sampling strategy [14], the optimal c		
CondConstraints.md	closed world constraints, neg_ratio=1	oui	N/A</td> </tr> <tr> <td>closed world constraints</td> <td>1</td> <td>801		
CondConstraints.md	no constraints, neg_ratio=1	oui	</thead> <tbody> <tr> <td>no constraints</td> <td>1</td> <td>667		
CondConstraints.md	open world constraints, neg_ratio=1	oui	647</td> </tr> <tr> <td>open world constraints</td> <td>1</td> <td>663		

Article	Entité	Trouvé	Extrait (tableau)		
dans.md	CAKE	oui	nerate informative negative triplets; CAKE [19]: a framework which leverages ext		
dans.md	Diversified and Adaptive Negative Sampling	oui	Xiv:2410.07592v2 [cs.AI] 26 Oct 2025 # Diversified and Adaptive Negative Sampling on Knowledge Graphs		
dans.md	KBGAN	oui	ous state-of-the-art approaches such as KBGAN [3], we adopt a generative adversarial		
dans.md	Markov Chain Monte Carlo Negative Sampling	NON	-		
dans.md	NSCaching	oui	nodes via random walks on the graph; NSCaching [39]: a model that employs importance		
dans.md	Popularity-weighted Negative Sampling	oui	edge graph sampled in a uniform [33] or popularity-weighted manner [17]. Although random sampling i		
dans.md	Random Negative Sampling	oui	ive triplets are often obtained through random sampling. More recent works [34, 3] exp		
dans.md	Self-adversarial Negative Sampling	oui	weighted sampling; Self-adv [28]: a self-adversarial negative sampling methodology; MCNS		
dans.md	SMILE	oui	triplets to sample negative triplets; SMILE [22]: a framework which employs speci		
dans.md	Structure-Aware Negative Sampling	oui	rent heuristics or learning strategies. Structure-aware models [1, 37] exploit the graph struct		
DeMix.md	Adaptive Denoising Mixup	oui	] 15 Oct 2023 # Negative Sampling with Adaptive Denoising Mixup for Knowledge Graph Embedding Xiangnan		
DeMix.md	Bernoulli Sampling	oui	riples from a uniform distribution. - <i>Bernoulli Sampling</i> [24]. which sample negative t		
DeMix.md	CANS	oui	d by utilizing the graph structure. - <i>CANS</i> [16]. The CANS is a component of CAKE		
DeMix.md	ESNS	oui	ainly compare CANS instead of CAKE. - <i>ESNS</i> [29]. It takes semantic similarities a		
DeMix.md	NSCaching	oui	negative triples with high scores, and NSCaching[31] introduces a caching mechanism to s		
DeMix.md	RW-SANS	oui	ing to the current embedding model. - <i>RW-SANS</i> [1]. It samples negative triples from		
DeMix.md	Self-adversarial Sampling	oui	.s. Epoch based on RotatE. Normal means self-adversarial negative sampling. Leakage means ensuri		
DeMix.md	Uniform Sampling	oui	$\mathbb{P}(e_j, y_j) \sim \begin{cases} \text{Uniform} \\ \mathcal{T}_{\cap} \end{cases} \& \text{if } (\mathbf{m}$		

Article	Entité	Trouv é	Extrait (tableau)		
DHNS.md	Bernoulli Negative Sampling	oui	ding to ambiguous training signals [4]. Bernoulli sampling [31] employs a Bernoulli distr		
DHNS.md	Diffusion-based Hierarchical Negative Sampling	oui	Xiv:2501.15393v1 [cs.AI] 26 Jan 2025 # Diffusion-based Hierarchical Negative Sampling for Multimodal Knowle		
DHNS.md	KBGAN	oui	nerate higher-quality negative triples. KBGAN [4] and IGAN [30] use Generative Advers		
DHNS.md	MANS	oui	lenging negatives from multimodal data. MANS [37] emphasizes modality-aware NS to al		
DHNS.md	MMRNS	oui	or generating diverse negative triples. MMRNS [34] introduces a relation-enhanced NS		
DHNS.md	NSCaching	oui	E models to distinguish from positives. NSCaching [40] utilizes additional memory to stor		
DHNS.md	SANS	oui	ality negative triples during training. SANS [1] leverages graph structure informati		
DHNS.md	Uniform Negative Sampling	oui	th MMKGC and NS performances, including uniform sampling (Uniform) [3], Bernoulli sampl		
DMNS.md	Conditional Diffusion-based Multi-level Negative Sampling	oui	om the latent space. Our method, called Conditional Diffusion-based Multi-level Negative Sampling (DMNS), leverages the		
DMNS.md	Dynamic Negative Sampling	oui	egative nodes, such as popularity [30], dynamic selections based on current predictions		
DMNS.md	KBGAN	oui	sampling. Among them, GraphGAN [46] and KBGAN [5] learns a distribution over negative		
DMNS.md	Markov Chain Monte Carlo Negative Sampling	NON	-		
DMNS.md	Popularity-based Negative Sampling	oui	to select hard negative nodes, such as popularity [30], dynamic selections based on curre		
DMNS.md	Uniform Negative Sampling	oui	ol the quality of negative nodes. While uniform negative sampling [14, 44] is simple, i		
DNS.md	Distributional Negative Sampling	oui	Xiv:1908.06178v1 [cs.LG] 16 Aug 2019 # Distributional Negative Sampling for Knowledge Base Co		
DNS.md	Random Negative Sampling	oui	whereas negative ones are generated by random sampling of entities. In this paper, we		
DomainNS.md	Domain Sampling	oui	relation can only belong to a specific domain. When the corrupted entity comes from o		
EANS.md	Entity Aware Negative Sampling	oui	Xiv:2210.06242v1 [cs.LG] 12 Oct 2022 # Entity Aware Negative Sampling with Auxiliary Loss o		
EANS.md	KB-GAN	oui	are the performances of the models with KB-GAN(Cai and Wang, 2017), NS-Caching(Zhang e		

Article	Entité	Trouv é	Extrait (tableau)		
EANS.md	NS-Caching	oui	models with KB-GAN(Cai and Wang, 2017), NS-Caching(Zhang et al., 2019), self-adversarial s		
EANS.md	SANS	oui	adversarial sampling(Sun et al., 2019), SANS(Ahrabian et al., 2020) and our EANS met		
EANS.md	Self-Adversarial Negative Sampling	oui	the graph. (Sun et al., 2019) proposed self-adversarial negative sampling, which give differenc		
EANS.md	Uniform Negative Sampling	oui	used because of its efficiency. Such a uniform random sampling method can be effective		
ERDNS.md	Bernoulli Sampling	oui	the performance of KGEs. For instance, Bernoulli sampling [29] assigns different probabi		
ERDNS.md	Entity-Relation Distribution-aware Negative Sampling	oui	# Entity-Relation Distribution-aware Negative Sampling for Knowledge Graph E		
ERDNS.md	KBGAN	oui	natural language processing (NLP) [7]. KBGAN [3], IGAN [26], and NSCaching [34] crea		
ERDNS.md	NSCaching	oui	ng (NLP) [7]. KBGAN [3], IGAN [26], and NSCaching [34] create a negative candidate set \$C		
ERDNS.md	Self-Adv	oui	egative samples in the loss function in Self-Adv [21] and SANS [1]. However, these appro		
ERDNS.md	Subsampling	oui	e the KGE performance. More recently, a Subsampling work [10] explores the hyperparameter t		
ERDNS.md	Uniform Sampling	oui	. Simple negative sampling methods like uniform sampling [2] generate negative samples		
eTruncatedUNS.md	Epsilon-Truncated Uniform Negative Sampling	NON	-		
GHN.md	Generative Hard Negative Mining	oui	proving Knowledge Graph Completion with Generative Hard Negative Mining **Zile Qiao ¹ , Wei Ye ² , Dingyao Yu ² , T		
GHN.md	GHN-SL	oui	WN18RR and FB15k-237 datasets. GHN-SL denotes the proposed GHN without self-i		
GHN.md	In-batch Negative Sampling	oui	roblem by contrasting carefully sampled in-batch, pre-batch and self negatives, inspirin		
GibbsNS.md	EANS	oui	egative sampling, KBGAN, NSCaching, and EANS [40], were used to make comparisons wit		
GibbsNS.md	Gibbs Negative Sampling	oui	earch offers a unique strategy based on Gibbs sampling and connection embedding to im		
GibbsNS.md	KBGAN	oui	is of static negative sampling methods, KBGAN [17] found that the majority of negativ		

Article	Entité	Trouv é	Extrait (tableau)		
GibbsNS.md	NSCaching	oui	e samples across various relationships. NSCaching [21] designed two modules: head entity		
GibbsNS.md	Uniform Random Negative Sampling	oui	Four negative sampling methods, namely, uniform random negative sampling, KBGAN, NSCaching, an		
GNDN.md	KBGAN	oui	ce a significant difference between the KBGAN model of [19] and KCGAN as proposed her		
GNDN.md	Knowledge Completion GAN	oui	s (GANs), we propose a novel framework, Knowledge Completion GANs (KCGANs), for competitively training ge		
GNS.md	Good Negative Sampling	oui	# Good negative sampling for triple classifica		
GNS.md	Pseudo Typed Negative Sampling	oui	ernoulli Negative Sampling (BNS) and 3) Pseudo Typed Negative Sampling (PTNS). UNS generates		
GNS.md	Uniform Negative Sampling	oui	mpling is composed of three methods: 1) Uniform Negative Sampling (UNS), 2) Bernoulli N		
HaSa.md	Hard InfoNCE	oui	s, lead to better performance [13, 15]. Hard InfoNCE samples negative tails from a distribut		
HaSa.md	Hardness and Structure-Aware Contrastive Knowledge Graph Embedding	oui	15 [cs.AI] arXiv:2305.10563v2 # HaSa: Hardness and Structure-Aware Contrastive Knowledge Graph Embedding **Honggen Zhan...		
HaSa.md	KBGAN	oui	quad (4) \$\$ Similar to RotatE [26] and KBGAN [4], we use the following negative samp		
HaSa.md	Self-adversarial negative sampling	oui	igher-quality negative samples, called <i>self-adversarial</i> or hard negative samples, lead to bett		
HaSa.md	Simple InfoNCE	oui	d, we only generate \$K\$ negative tails. Simple InfoNCE[22] samples negative tails from a distr		
HTENS.md	Hierarchical Type Enhanced Negative Sampling	oui	# Hierarchical Type Enhanced Negative Sampling for Knowledge Graph E		
HTENS.md	KBGAN(pre)	oui	The GAN-based sampling methods, such as KBGAN[3] and IGAN[13], set the KGE model to b		
HTENS.md	Uniform Sampling	oui	ibution based sampling methods, such as Uniform sampling[2] and Bernoulli sampling[14],		
KBGAN.md	KBGAN	oui	Xiv:1711.04071v3 [cs.CL] 16 Apr 2018 # KBGAN: Adversarial Learning for Knowledge Gra		
KBGAN.md	Uniform Negative Sampling	oui	tail entity, and randomly sample from a uniform distribution (Bordes et al., 2013). Unf		
KSGAN.md	KBGAN	oui	ed by an adversarial learning framework KBGAN, this paper proposes a new knowledge se		

Article	Entité	Trouv é	Extrait (tableau)				
KSGAN.md	Knowledge Selective Adversarial Network	oui	# A Knowledge Selective Adversarial Network for Link Prediction in Knowledge Graph				
KSGAN.md	KSGAN(ComplEx+Trans E)	oui	selective adversarial network, named as KSGAN, using a knowledge selector for high-qu				
LAS.md	KBGAN	oui	h outperforms GAN-based sampling method KBGAN. INDEX TERMS Knowledge graph, know				
LAS.md	Loss Adaptive Sampling	oui	nowledge representation learning, named loss adaptive sampling (LAS) mechanism, which is effi				
LAS.md	Uniform Random Negative Sampling	oui	ses in training FB15k237 by TransE with uniform random negative triplets selection. To allevi				
LEMON.md	Adversarial Language Model-driven Negative Sampling	NON	-				
LEMON.md	Adversarial Uniform Negative Sampling	oui	d hop size $H = 3$. Using non adversarial/uniform negative sampling the number of negativ				
LEMON.md	KBGAN	oui	1)</td> </tr> <tr> <td>KBGAN [6]</td> <td> $O(t)$ </				
LEMON.md	Language Model-driven Negative Sampling	oui	rXiv:2203.04703v1 [cs.AI] 9 Mar 2022 # Language Model-driven Negative Sampling **Mirza Mohtashim Al				
LEMON.md	NSCaching	oui	t)</td> </tr> <tr> <td>NSCaching [22]</td> <td> $O(1)$ <				
LEMON.md	Self-Adv. RW-SANS	oui	\	)</td> </tr> <tr> <td>Self-Adv. RW-SANS [2]</td> <td> $O(r)$ mat			
LEMON.md	Self-Adv. SANS	oui	2)</td> </tr> <tr> <td>Self-Adv. SANS [2]</td> <td> $O(\backslash$	\mat			
LEMON.md	Uniform Negative Sampling	oui	21, 8, 3]. One of the early works named Uniform [10] proposed a fixed distribution and				
LEMON.md	Uniform RW-SANS	oui	2)</td> </tr> <tr> <td>Uniform RW-SANS [2]</td> <td> $O(r)$ mat				
LEMON.md	Uniform SANS	oui	}\	\$</td> </tr> <tr> <td>Uniform SANS [2]</td> <td> $O(\backslash$	\mat		
Localcognitive.md	Local-cognitive negative sampling	oui	tion patterns. We also propose a novel local-cognitive negative sampling method, by integratin				
Localcognitive.md	Self-adversarial negative sampling	oui	sed to utilize both prior knowledge and self-adversarial learning for effective optimization. Ex				
LTS.md	Bernoulli Negative Sampling	oui	be divided into uniform distributions, Bernoulli distributions and improved Bernoulli di				

Article	Entité	Trouv é	Extrait (tableau)		
LTS.md	Negative Sampling of Potential Topic Similarities	oui	ge Alignment Based on Negative Sampling of Potential Topic Similarities *Wei Tian School of Computer, Zhongyua		
LTS.md	Uniform Negative Sampling	oui	tatic distributions can be divided into uniform distributions, Bernoulli distributions		
M2ixKG.md	KBGAN	oui	tribution. IGAN [Wang et al., 2018] and KBGAN [Cai and Wang, 2018] introduce generati		
M2ixKG.md	MixKG HNM-CES	NON	-		
M2ixKG.md	MixKG HNM-SF	NON	-		
M2ixKG.md	NSCaching	oui	ct high-quality negative samples, while NSCaching [Zhang et al., 2019] utilizes cached-ba		
M2ixKG.md	RW-SANS	oui	ples from the \$\$-hop neighborhood. * RW-SANS [Ahrabian et al., 2020] is similar		
M2ixKG.md	SANS	oui	-quality negative samples. Differently, SANS [Ahrabian et al., 2020] absorbs graph s		
M2ixKG.md	Uniform Negative Sampling	oui	. Most current KG embedding models use uniform sampling to generate false triplets [Bo		
MCNS.md	Dynamic Negative Sampling	oui	rm sampling for negative samples. Then, dynamic negative sampling (DNS) [41] is propos		
MCNS.md	IRGAN	oui	0]). * GAN-based Negative Sampling . IRGAN [35] trains generative adversarial nets		
MCNS.md	KBGAN	oui	ptively generate hard negative samples. KBGAN [3] employs a sampling-based adaptation		
MCNS.md	Markov chain Monte Carlo Negative Sampling	oui	scalable negative sampling strategy, Markov chain Monte Carlo Negative Sampling (MCNS), which appli		
MCNS.md	PinSAGE	oui	x (DNS [41]) and personalized PageRank (PinSAGE [40]). * **GAN-based Negative Sampling .		
MCNS.md	Power of Degree	oui	ng distribution proportional to the 3/4 power of degree by tuning the power parameter. Most lat		
MCNS.md	Random Negative Sampling	oui	sample positive node pairs, such as by random walk [27], the second-order proximity [		
MCNS.md	Weighted Approximate-Rank Pairwise	oui	the hard negative samples by rejection (WARP [44]), sampling-max (DNS [41]) and pers		
MCNS.md	Weighted Regularized Matrix Factorization	oui	clude Power of Degree [24], RNS [4] and WRMF [15], which can be summarized as \$p_n(v		
MDNCaching.md	Bernoulli Negative Sampling	oui	e introduced various strategies such as Bernoulli negative sampling to generate quality n		

Article	Entité	Trouv é	Extrait (tableau)		
MDNCaching.md	KBGAN	oui	ative sampling strategies IGAN [12] and KBGAN [3] were introduced to generate negativ		
MDNCaching.md	Matrix Decomposed Negative Caching	oui	duces a new strategy called MDNCaching (Matrix Decomposed Negative Caching), which generates negatives considering		
MDNCaching.md	NSCaching	oui	sibility of generating false negatives. NSCaching [17] was proposed to overcome the chall		
MDNCaching.md	Structure Aware Negative Sampling	oui	ined, which adds extra costs. Recently, Structure Aware Negative Sampling (SANS) [1] strategy w		
NMiss.md	Corrupting Positive Instances	oui	nts in the following sections. ### 3.2 Corrupting positive instances : C We use a method described in [24]		
NMiss.md	Near Miss Sampling	oui	eighbor sampling for 5 epochs. ### 3.6 Near Miss sampling : nmiss The nearest neighbor		
NoiGAN.md	CKRL (40%)	oui	19)), (3) noise aware KGE models (e.g., CKRL (Xie et al., 2018)) and (4) KGE models		
NoiGAN.md	KBGAN	oui	18)) and (4) KGE models with GAN (e.g., KBGAN (Cai & Wang, 2017)). In particular, the		
NoiGAN.md	NoiGAN	oui	w as a conference paper at ICLR 2020 # NOIGAN: NOISE AWARE KNOWLEDGE GRAPH EMBEDDING		
NSCaching.md	Bernoulli Negative Sampling	oui	rmance. A better sampling scheme, i.e., Bernoulli sampling, is introduced in [42]. It imp		
NSCaching.md	IGAN	oui	y, there are two pioneered works, i.e., IGAN [39] and KBGAN [9], attempting to addre		
NSCaching.md	KBGAN + scratch	oui	wo pioneered works, i.e., IGAN [39] and KBGAN [9], attempting to address these challe		
NSCaching.md	NSCaching	oui	Xiv:1812.06410v2 [cs.AI] 18 Jan 2019 # NSCaching: Simple and Efficient Negative Sampling		
NSCaching.md	Uniform Negative Sampling	oui	Due to its simplicity and efficiency, uniform sampling is broadly used in KG embeddin		
PNS.md	Bernoulli Negative Sampling	oui	> </tr> <tr> <td>TransH(bern)</td> <td>401</td> <td>		
PNS.md	Probabilistic Negative Sampling	oui	nhancing Knowledge Graph Embedding with Probabilistic Negative Sampling Citation in BibTeX f		
RAAKGC.md	In-Batch Negative Sampling	oui	are the self-negative samples (SN) and in-batch negative samples (IBN) (Wang et al. 202		

Article	Entité	Trouvé	Extrait (tableau)		
RAAKGC.md	Relation-Aware Anchor Enhancement	oui	2025 # Knowledge Graph Completion with Relation-Aware Anchor Enhancement Duanyang Yuan ¹ , Sihang Zhou ¹ , Xiao		
RCWC.md	Naive Negative Sampling	NON	-		
RCWC.md	Range-Constrained with Co-occurrence Negative Sampling	oui	hold, δ_{rcwc} . As a result, the range-constrained with co-occurrence (rcwc) negative samp		
RCWC.md	Self-Adversarial Negative Sampling	oui	ce the complexity of negative sampling, self-adversarial training was adopted in RotatE [Sun et		
ReasonKGE.md	Random Negative Sampling	oui	ve triples is done either completely at random [9], relying on the (local) closed worl		
ReasonKGE.md	ReasonKGE	oui	we present a novel iterative approach <i>ReasonKGE</i> that identifies dynamically via symbol		
ReasonKGE.md	Static Sampling	oui	Therefore, instead of pre-computing a static set of negative examples, we propose to		
RUGA.md	soft Rules and graph adversarial learning	oui	rove the completion of knowledge graph: soft Rules and graph adversarial learning (RUGA). Firstly, the traditional know...		
SANS.md	KBGAN	oui	egative sampling (Bordes et al., 2013), KBGAN (Cai and Wang, 2018), and NSCaching (Zh		
SANS.md	NSCaching	oui	train compared to GAN-based methods is NSCaching (Zhang et al., 2019), which involves us		
SANS.md	Self-Adversarial Negative Sampling	oui	sy negatives, Sun et al. (2019) propose Self-Adversarial negative sampling, which weighs each sa		
SANS.md	Self-Adversarial Random Walk Structure Aware Negative Sampling	NON	-		
SANS.md	Self-Adversarial Structure Aware Negative Sampling	NON	-		
SANS.md	Uniform Negative Sampling	oui	o simple corruption distributions, i.e. uniform, yielding easy uninformative negatives		
SANS.md	Uniform Random Walk Structure Aware Negative Sampling	NON	-		
SANS.md	Uniform Structure Aware Negative Sampling	NON	-		
SelfAdv.md	KBGAN	oui	en from (Cai (Cai & Wang, 2017), where KBGAN uses a ComplEx negative sample generato		
SelfAdv.md	Self-Adversarial Negative Sampling	oui	space. In addition, we propose a novel self-adversarial negative sampling technique for efficie		
SelfAdv.md	Uniform Negative Sampling	oui	loss samples the negative triplets in a uniform way. Such a uniform negative sampling s		
SNS.md	KBGAN	oui	trained to learn embeddings. IGAN [8], KBGAN [3], KSGAN [7] are three existing GAN-b		

Article	Entité	Trouv é	Extrait (tableau)		
SNS.md	NSCaching	oui	ive adversarial network(GAN)-based, and NSCaching, structure aware negative sampling(SANS		
SNS.md	Simple Negative Sampling	oui	science logo](page_1_image_1_v2.jpg) # Simple negative sampling for link prediction i		
SNS.md	Uniform Random Negative Sampling	oui	triples. To the best of our knowledge, uniform-random, generative adversarial network(GAN)-ba		
SparseNSG.md	Negative Sampling Based on Frequency of Relational Association Entities	oui	# A Novel Negative Sampling Based on Frequency of Relational Association Entities for Knowledge Graph Embedding Yi Zhang		
STC.md	Soft Type Constraint	oui	se a method for negative sampling named Soft Type Constraint (STC). STC improves the probability of		
TANS.md	Self-Adversarial Negative Sampling	oui	in KGE relies on smoothing methods like Self-Adversarial Negative Sampling (SANS) and subsamplin		
TANS.md	Self-Adversarial Negative Sampling w/ Base	oui	</th> <th>SANS</th> <th>SANS w/ Base</th> <th>SANS w/ Freq</th>		
TANS.md	Self-Adversarial Negative Sampling w/ Freq	oui	<th>SANS w/ Base</th> <th>SANS w/ Freq</th> <th>SANS w/ Uniq</th>		
TANS.md	Self-Adversarial Negative Sampling w/ Uniq	oui	<th>SANS w/ Freq</th> <th>SANS w/ Uniq</th> </tr> <tr> <th col		
TANS.md	Triplet Adaptive Negative Sampling	oui	loss in KGE and induces a new NS loss, Triplet Adaptive Negative Sampling (TANS), that can cove		
TANS.md	Triplet Adaptive Negative Sampling w/ Base	oui	</th> <th>TANS</th> <th>TANS w/ Base</th> <th>TANS w/ Freq</th>		
TANS.md	Triplet Adaptive Negative Sampling w/ Freq	oui	<th>TANS w/ Base</th> <th>TANS w/ Freq</th> <th>TANS w/ Uniq</th>		
TANS.md	Triplet Adaptive Negative Sampling w/ Uniq	oui	<th>TANS w/ Freq</th> <th>TANS w/ Uniq</th> </tr> </thead> <tbody>		
TANS.md	Uniform Negative Sampling	oui	om word embedding to KGE with utilizing uniform distribution as its noise distribution.		
TANS.md	Uniform Negative Sampling w/ Base	NON	-		
TANS.md	Uniform Negative Sampling w/ Freq	NON	-		
TANS.md	Uniform Negative Sampling w/ Uniq	NON	-		
TruncatedNS.md	Truncated Negative Sampling	oui	ing Attribute Character Embeddings with Truncated Negative Sampling for Entity Alignment		

Article	Entité	Trouvé	Extrait (tableau)		
TuckerDNCaching.md	Bernoulli Negative Sampling	oui	e fact (<i>Michelangelo, Gender, Male</i>). Bernoulli negative sampling (Wang et al., 2014) i		
TuckerDNCaching.md	KBGAN	oui	ributions, IGAN (Wang et al., 2018) and KBGAN (Cai et al., 2018) were introduced as n		
TuckerDNCaching.md	KSGAN	oui	e embedding model as the discriminator. KSGAN (Hu et al., 2019) is an extension of KB		
TuckerDNCaching.md	MDNCaching	oui	method by extending the previous work, MDNCaching (Madushanka et al., 2022) by introduc		
TuckerDNCaching.md	NSCaching	oui	h the aim of generating hard negatives, NSCaching (Zhang et al., 2019) introduces an appr		
TuckerDNCaching.md	TuckerDNCaching	oui	for updates](page_1_image_1_v2.jpg) # TuckerDNCaching: high-quality negative sampling with tu		
TypeConstraints.md	Local Closed-World Assumption	oui	cases, it can be beneficial to apply a local closed-world assumption that approximates the semantics of rela		
TypeConstraints.md	Type-Constrained Negative Sampling	oui	Xiv:1508.02593v2 [cs.AI] 28 Aug 2015 # Type-Constrained Representation Learning in Knowledge Gr		

B. Preuve - MENTIONNÉS seulement (134)

Article	Entité	Trouvé	Extrait (prose)
AdaptativeNS.md	Adversarial Negative Sampling	oui	s function, facilitating more effective adversarial learning. Unlike the traditional TransE
AdaptativeNS.md	Frequency-based Negative Sampling	oui	t selects negative samples based on the frequency characteristics of positive samples. Th
AdaptativeNS.md	Random Negative Sampling	oui	nce, conventional methods have employed random negative sampling to augment the model'
ADNS.md	Bernoulli Negative Sampling	oui	egative samples. In the training phase, Bernoulli sampling [10] is used to decide if the
ADNS.md	Near Miss Sampling	oui	al Sampling, Nearest Neighbor sampling, Near Miss sampling. Nearest Neighbor sampling and
ADNS.md	Nearest Neighbor Sampling	oui	scussed in [7] are Relational Sampling, Nearest Neighbor sampling, Near Miss sampling. Nearest N
ADNS.md	Relational Sampling	oui	e other approaches discussed in [7] are Relational Sampling, Nearest Neighbor sampling, Ne
ADNS.md	Typed Sampling	oui	pool to corrupt positive instances [7]. Typed Sampling is the negative sampling based

Article	Entité	Trouvé	Extrait (prose)
ASA.md	Self-adversarial Negative Sampling	oui	are Graph Attention Model With Adaptive Self-adversarial Training *Xiao Qin ¹ l, Nasrullah Shei
BatchNS.md	Uniform Negative Sampling	oui	lion users vs. 1 million products. With uniform negative sampling over all nodes, the l
CAKE.md	KBGAN	oui	2014). (2) Adversarial-based sampling: KBGAN (Cai and Wang, 2018) integrates the KGE
CAKE.md	Type-Constrained Negative Sampling	oui	omain-based NS (Wang et al., 2019b) and type-constrained NS (Denis et al., 2015) both leverage d
CANS.md	Bernoulli Negative Sampling	oui	ion, the L1 norm was employed, and the "bern" [10] strategy was adopted, to decide w
CANS.md	Uniform Negative Sampling	oui	ugh effective at detecting noises, with uniform negative sampling methods, and a harsh
CCS.md	Uniform Random Sampling	oui	works, such as the TransE model, use a uniform random sampling method to generate negative sa
ConceptDriven.md	Entity-Aware Negative Sampling	oui	high-quality negatives. [21] proposes a entity-aware NS method, which tries to sample negati
ConceptDriven.md	Reinforcement Learning Negative Sampling	oui	lops a novel NS model, which works as a reinforcement learning (RL) agent to explore high-quality nega
CondConstraints.md	Bernoulli Negative Sampling	oui	for TransH and is sometimes called the Bernoulli trick [15]. The Bernoulli trick involve
CondConstraints.md	Conditional Constraint-Based Negative Sampling	oui	CEUR-WS.org/Vol-2635/paper3.pdf # Conditional Constraints for Knowledge Graph Embeddings Michael
CondConstraints.md	Filtered Negative Sampling	oui	ransE embedding model: using so-called <i>filtered</i> negative samples [1]. Filtered negativ
CondConstraints.md	Locally Closed World Negative Sampling	oui	ail with another entity) or to assume a locally closed world in which any valid triple entails a who
dans.md	GNDN	oui	as KBGAN [3], IGAN [30], HeGAN [11] and GNDN [38], which learn the underlying sample
dans.md	HeGAN	oui	GAN) [10] such as KBGAN [3], IGAN [30], HeGAN [11] and GNDN [38], which learn the und
dans.md	IGAN	oui	rial nets (GAN) [10] such as KBGAN [3], IGAN [30], HeGAN [11] and GNDN [38], which l
DeMix.md	IGAN	oui	high scores. For example, KBGAN[6] and IGAN[23] introduce a generative adversarial
DeMix.md	KBGAN	oui	triples with high scores. For example, KBGAN[6] and IGAN[23] introduce a generative
DeMix.md	MixKG	oui	harder negative triples for KGE such as MixKG [8]. Different from existing methods, D

Article	Entité	Trouvé	Extrait (prose)
DHNS.md	DMNS	oui	original data. In graph-structured data, DMNS[21] employs DDPM to generate negative n
DHNS.md	IGAN	oui	quality negative triples. KBGAN [4] and IGAN [30] use Generative Adversarial Network
DHNS.md	NS-KGE	oui	ing (NSCach) [40], KBGAN [4], SANS [1], NS-KGE [13], MANS [37] and MMRNS [34]. **Impl
DMNS.md	MixGCF	oui	ve examples from k -hop neighborhoods. MixGCF [20] synthesizes hard negative examples
DMNS.md	Subgraph-based Negative Sampling	oui	ts the representations of the enclosing subgraphs. For example, SEAL [53] proposes the us
DNS.md	Bernoulli Sampling	oui	tance. For each triple (h, r, t) , we use bernoulli sampling (as introduced by Wang et al.
DNS.md	Domain Sampling	oui	s to RNS. Xie et al. [2017] introduce a domain sampling approach, where for each relat
DNS.md	Nearest Neighbor Sampling	oui	tandard KBC model is trained which uses nearest neighbor sampling (on the frozen model in the fi
DNS.md	Typed Sampling	oui	eps, first a KBC model is trained using Typed-Sampling i.e. corrupted entities belong
DomainNS.md	Bernoulli Negative Sampling	oui	h SGD until convergence. We employ the “Bernoulli” sampling method to generate incorrect
DomainNS.md	Uniform Negative Sampling	NON	-
ERDNS.md	IGAN	oui	nguage processing (NLP) [7]. KBGAN [3], IGAN [26], and NSCaching [34] create a negat
ERDNS.md	Non-Sampling	oui	false negative samples (i.e., true but non-observed triples) for the latter. We di
ERDNS.md	SANS	oui	the loss function in Self-Adv [21] and SANS [1]. However, these approaches primaril
eTruncatedUNS.md	Uniform Negative Sampling	oui	les, we propose an ϵ -truncated uniform negative sampling method. We also swap
GHN.md	Adversarial Contrast	NON	-
GHN.md	Approximate Nearest Neighbor Negative Sampling	oui	of generated negatives, we construct an approximate nearest neighbor negative sampling strategy (ANN) inspir
GHN.md	Hard Negative Mixing	NON	-
GHN.md	NSCaching	NON	-
GHN.md	Pre-batch Negative Sampling	oui	contrasting carefully sampled in-batch, pre-batch and self negatives, inspiring us to lev
GHN.md	Reinforced Negative Sampling	NON	-

Article	Entité	Trouvé	Extrait (prose)
GHN.md	Self Negative Sampling	oui	chor and thus enjoying good hardness. A self-information-enhanced contrasting strate
GNDN.md	Random Negative Sampling	oui	rator is used to generate an image from random noise and the discriminator is employed
GNS.md	Bernoulli Negative Sampling	oui	ound ontological knowledge; 2) in turn, Bernoulli's distribution is used for decreasing t
GNS.md	Iterative Negative Sampling	oui	ity to corrupt. ReasonKGE [9] with its Iterative Negative Sampling (INS) identifies dyna
GNS.md	ReasonKGE	oui	o pick head or tail entity to corrupt. ReasonKGE [9] with its Iterative Negative Samplin
GNS.md	Typed Negative Sampling	oui	f ontologies in the negative sampling. Typed Negative Sampling (TNS) [10] which is s
HaSa.md	Noise contrastive estimation	oui	s proposed by [20], which is similar to noise contrastive estimation (NCE) [11]. More recently, InfoNCE loss
HTENS.md	Bernoulli Negative Sampling	oui	ethods, such as Uniform sampling[2] and Bernoulli sampling[14], assign fixed sampling pro
HTENS.md	Cache-based Negative Sampling	oui	ator to construct negative samples. The cache-based sampling method[17] maintains a h
HTENS.md	Hard Type Constraint	oui	es that are more difficult to classify, Hard Type Constraint[9] and Soft Type Constraint[15] are pro
HTENS.md	IGAN	oui	sampling methods, such as KBGAN[3] and IGAN[13], set the KGE model to be trained as
HTENS.md	Soft Type Constraint	oui	o classify, Hard Type Constraint[9] and Soft Type Constraint[15] are proposed. However, as the train
IGAN.md	Bernoulli Negative Sampling	oui	or tail with equal probability; (2) “ bern ” method from (Wang et al. 2014) which
IGAN.md	Random Negative Sampling	oui	ks construct negative samples through a random mode, by which the samples are often to
IGAN.md	Uniform Negative Sampling	oui	to replace head or tail entity, (1) “ unif ” method which replaces head or tail w
KBGAN.md	Bernoulli Negative Sampling	oui	nal complexity. Besides, we adopt the “bern” sampling technique (Wang et al., 2014)
KBGAN.md	Type-constrained Negative Sampling	NON	-
KSGAN.md	Bernoulli Negative Sampling	oui	om a given true triple (h, r, t) with Bernoulli distribution. The aforementioned negati
KSGAN.md	IGAN	oui	ediction task. In contrast to KBGAN and IGAN [11], KSGAN leverages a knowledge selec
KSGAN.md	Uniform Negative Sampling	NON	-

Article	Entité	Trouvé	Extrait (prose)
LAS.md	Bernoulli Negative Sampling	oui	f $((h, r, t), (h'_s, r, t'_s))$ with a Bernoulli distribution $B(1, p_{up})$. We consid
LAS.md	Domain Sampling	oui	e, corrupted triplets which violate the domain/range restriction of its relation is de
LAS.md	Near-miss Sampling	oui	iminator, and it can be regarded as the near miss sampling in [26]. In this part of exper
LAS.md	Nearest-neighbour Sampling	oui	], the static negative sampling models, nearest-neighbour sampling and near-miss sampling, are pr
LAS.md	Type-constrained Negative Sampling	NON	-
LEMON.md	Adversarial Negative Sampling	oui	ed distribution [7]. In [9], Generative Adversarial networks (GANs) are used for generating
LEMON.md	Bernoulli Negative Sampling	oui	ties to be corrupted [12, 14]. In [20], Bernoulli distribution is considered to limit the
LEMON.md	Structure-aware Negative Sampling	oui	nerative Adversarial Network-based, and Structure-aware Approaches. **Uniform Distribution-bas
Localcognitive.md	KBGAN	oui	effectively learn structured knowledge. KBGAN (Cai and Wang, 2018) uses knowledge gra
Localcognitive.md	Type-constraint training	oui	egative sampling method, by integrating type-constraint training technique (Krompaß et al., 2015) with s
Localcognitive.md	Uniform negative sampling	oui	$\mathcal{E}^{(tr)}$. To achieve this, we conduct a uniform sampling individually in $\mathcal{E}^{(tr)}$
LTS.md	Custom Pooled Sampling	oui	ling, Dynamic Distribution Sampling and Custom Pooled Sampling respectively. Static distribut
LTS.md	Dynamic Distribution Sampling	oui	These are Static Distribution Sampling, Dynamic Distribution Sampling and Custom Pooled Sampling res
LTS.md	Random Negative Sampling	oui	o solve the zero-loss problem caused by random negative sampling. Entity alignment al
LTS.md	Static Distribution Sampling	oui	igh quality negative triplet. These are Static Distribution Sampling, Dynamic Distribution Sampling
M2ixKG.md	Bernoulli Negative Sampling	oui	form sampling [Bordes et al., 2013] and Bernoulli sampling [Wang et al., 2014]. However,
M2ixKG.md	IGAN	oui	samples from a dynamical distribution. IGAN [Wang et al., 2018] and KBGAN [Cai and
M2ixKG.md	MixKG	oui	Xiv:2202.09606v1 [cs.AI] 19 Feb 2022 # MixKG: Mixing for harder negative samples in

Article	Entité	Trouvé	Extrait (prose)
MCNS.md	Bayesian Personalized Ranking	oui	studied extensively in recommendation. Bayesian Personalized Ranking [29] proposes uniform sampling for nega
MDNCaching.md	IGAN	oui	GAN) based negative sampling strategies IGAN [12] and KBGAN [3] were introduced to g
MDNCaching.md	Uniform Negative Sampling	oui	existing negative sampling strategies, Uniform negative sampling is a widely used stra
NMiss.md	Importance Sampling	oui	e sampling was first proposed in [1] as importance sampling. A sampling solution that was
NMiss.md	Nearest Neighbor Sampling	oui	ber \$n_{s}\$ of negative samples. ### 3.5 Nearest Neighbor sampling : NN Most negative sampling m
NMiss.md	Noise Contrastive Estimation	oui	introduced by [16], who built upon the noise-contrastive estimation [10]. In these approaches negative samp
NMiss.md	Random Sampling	oui	ompletion, or link prediction. ### 3.1 Random sampling : R The simplest form of samp
NMiss.md	Relational Sampling	oui	terminated types as in Freebase. ### 3.4 Relational Sampling : REL Although typed or corru
NMiss.md	Typed Sampling	oui	mly produced negative samples. ### 3.3 Typed Sampling : T Knowledge graphs such as
NoiGAN.md	Random Negative Sampling	oui	r, negative samples generated through a random mode are often too easy to be discrimin
NSCaching.md	Self-paced GAN	oui	[39] and KBGAN [9] for KG embedding and self-paced GAN [12] for network embedding. # VI. CONC
RAAKGC.md	In-Batch-Relation Negative Sampling	oui	he same. Therefore, we also incorporate in-batch-relation negative samples (IBRN). This ![Framew
RAAKGC.md	Self-Negative Sampling	oui	in KG, and the negative samples are the self-negative samples (SN) and in-batch nega
RCWC.md	Bernoulli Negative Sampling	oui	ility of corrupting heads or tails as a Bernoulli distribution to avoid false negatives.
RCWC.md	Local-Closed World Assumption Negative Sampling	oui	et al. [2015], where it was called the local-closed world assumption (LCWA). LCWA assumes that head and tail
ReasonKGE.md	Distributional Negative Sampling	oui	d assumption [27]. Alternatives include Distributional Negative Sampling (DNS) [12] and its va
ReasonKGE.md	Near Miss Sampling	oui	approach [30]. <i>Nearest Neighbor</i> and <i>Near Miss sampling</i> [21] resp. exploit a pre-trai
ReasonKGE.md	Nearest Neighbor Sampling	oui	n or triple corruption approach [30]. <i>Nearest Neighbor</i> and <i>Near Miss sampling</i> [21] resp. ex

Article	Entité	Trouvé	Extrait (prose)
ReasonKGE.md	NSCaching	oui	samples from a node's neighborhood. The NSCaching sampling method [40] suggests to sample
ReasonKGE.md	Structure-Aware Negative Sampling	oui	egative sampling. The work [1] presents structure-aware negative sampling (SANS), which utilize
ReasonKGE.md	Type-Constrained Negative Sampling	oui	nother related method is concerned with type-constrained negative sampling [22]. Given a triple
RUGA.md	KBGAN	oui	herefore, Liwei Cai et al.[16] proposed KBGAN to solve the problem that negative samp
RUGA.md	Uniform Negative Sampling	oui	e of positive examples. The traditional uniform negative sampling method fixed the rati
SANS.md	Noise Contrastive Estimation	oui	andscape as simple random sampling-e.g. Noise Contrastive Estimation (NCE) (Gutmann and Hyvärinen, 2010)-pro
SANS.md	Random Walk Structure Aware Negative Sampling	NON	-
SANS.md	Structure Aware Negative Sampling	oui	rXiv:2009.11355v2 [cs.LG] 7 Oct 2020 # Structure Aware Negative Sampling in Knowledge Graphs
SNS.md	IGAN	oui	or part is trained to learn embeddings. IGAN [8], KBGAN [3], KSGAN [7] are three exi
SNS.md	KSGAN	oui	learn embeddings. IGAN [8], KBGAN [3], KSGAN [7] are three existing GAN-based sampli
SNS.md	Structure Aware Negative Sampling	oui	rial network(GAN)-based, and NSCaching, structure aware negative sampling(SANS) are four negati
SparseNSG.md	Bernoulli Negative Sampling	NON	-
STC.md	Bernoulli Negative Sampling	oui	lement two strategies (i.e. "unif" and "bern") for replacing head or tail with equal
STC.md	Uniform Negative Sampling	oui	4b], we implement two strategies (i.e. "unif" and "bern") for replacing head or tail
TANS.md	Base subsampling	NON	-
TANS.md	Frequency-based subsampling	oui	ling of Sun et al. (2019) (Base), their frequency-based subsampling (Freq) and unique-based subsampling (Un
TANS.md	Unique-based subsampling	oui	frequency-based subsampling (Freq) and unique-based subsampling (Uniq) for KGE. Kamigaito and Hayashi (
TruncatedNS.md	Random Negative Sampling	oui	the triple of positive examples with a random entity. The characteristics of structu
TuckerDNCaching.md	Entity Similarity-based Negative Sampling	oui	th large gradients in head/tail caches. Entity Similarity-based Negative Sampling (ESNS) (Yao et
TuckerDNCaching.md	ExtremeSelectCaching	oui	negative sampling method. We called it "ExtremeSelectCaching" as it evaluates the importance of all

Article	Entité	Trouvé	Extrait (prose)
TuckerDNCaching.md	IGAN	oui	14) for modeling dynamic distributions, IGAN (Wang et al., 2018) and KBGAN (Cai et a
TuckerDNCaching.md	Self-Adversarial Negative Sampling	oui	nerative adversarial networks (GAN) and self-adversarial methods have recently been utilized in
TuckerDNCaching.md	Structure Aware Negative Sampling	oui	ignores non-semantic similar neighbors, Structure Aware Negative Sampling (SANS) (Ahrabian et a
TuckerDNCaching.md	Uniform Negative Sampling	oui	simplicity and efficiency. For example, Uniform negative sampling (Bordes et al., 2013)
TypeAugmented.md	Bernoulli negative sampling	oui	ples, Wang et al.¹³ define a Bernoulli distribution to replace the head and ta
TypeAugmented.md	CAKE	oui	esides, Niu et al.³³ propose CAKE which automatically extract commonsense
TypeAugmented.md	Self-adversarial negative sampling	oui	ntly, Sun et al.¹⁵ propose a self-adversarial negative sampling strategy and design a
TypeAugmented.md	Soft Type Constraint	oui	proved negative sampling strategy named Soft Type Constraint (STC). It selects entities in the same
TypeAugmented.md	Type-constrained negative sampling	oui	istinct relations. We also design a new type-constrained negative sampling strategy to construct
TypeAugmented.md	Uniform negative sampling	oui	f the existing KEG models carry out the uniform sampling scheme as in¹². To

C1. Candidats faux négatifs - ÉVALUÉS (43)

Article	Entité	Où	Extrait		
CAKE.md	CANS	prose+table	he commonsense-aware negative sampling (CANS) module and the multi-view link predict		
CANS.md	CKRL (40%)	prose+table	ware knowledge representation learning (CKRL) framework was proposed, to incorporate		
ConceptDriven.md	CANS	prose+table	e, commonsense-aware negative sampling (CANS) module and the multi-view link predict		
dans.md	SANS	prose+table	hood of positive examples. For example, SANS [1] hypothesizes that entities that are		
DeMix.md	SANS	prose+table	to the current embedding model. - RW-SANS [1]. It samples negative triples from		
DeMix.md	Self-Adv	table	> </tr> <tr><td>TransE+Self-Adv</td><td>0.215^{Δ}</td></tr>		
DeMix.md	Commonsense-Aware Negative Sampling	prose+table	d by utilizing the graph structure. - CANS [16]. The CANS is a component of CAKE		

Article	Entité	Où	Extrait		
DHNS.md	Self-adversarial Negative Sampling	prose+table	ality negative triples during training. SANS [1] leverages graph structure informati		
DMNS.md	DMNS	prose+table	on-based Multi-level Negative Sampling (DMNS), leverages the Markov chain property o		
DomainNS.md	Frequency-based Negative Sampling	table	> <th>Relation</th> <th>Frequency</th> <th>Log(Frequency)</th>		
DomainNS.md	Random Negative Sampling	table	7.0</td> </tr> <tr> <td>Random Walk (Wei et al., 2016)</td> <t		
EANS.md	EANS	prose+table	called Entity Aware Negative Sampling (EANS), which is able to sample negative enti		
EANS.md	Structure-Aware Negative Sampling	prose+table	adversarial sampling(Sun et al., 2019), SANS(Ahrabian et al., 2020) and our EANS met		
ERDNS.md	Frequency-based Negative Sampling	table	<th>Sorted ER ID</th> <th>Frequency</th> </tr> </thead> <tbody>		
GibbsNS.md	Uniform Negative Sampling	table	<td rowspan="5">TransE</td> <td>Uniform</td> <td>357</td> <td>0		
GibbsNS.md	Entity-Aware Negative Sampling	prose+table	egative sampling, KBGAN, NSCaching, and EANS [40], were used to make comparisons wit		
GNDN.md	Frequency-based Negative Sampling	table	<td>Probability</td> <td>Frequency</td> </tr> <tr> <td>0.0		
LEMON.md	RW-SANS	table	</tr> <tr> <td>Uniform RW-SANS [2]</td> <td> $\mathcal{O}(\lvert \mat$		
LEMON.md	SANS	prose+table	ture-aware negative sampling technique (SANS) was proposed [2] to consider the neigh		
LEMON.md	Self-Adv	table	\	)\$</td> </tr> <tr> <td>Self-Adv. [15]</td> <td> $\mathcal{O}(\lvert$	\m
M2ixKG.md	Random Negative Sampling	table	h> <th>Uniform</th> <th>Random Mix</th> <th>HNMF-SF</th>		
M2ixKG.md	Uniform RW-SANS	prose+table	s of ComplEx for Uniform, Uniform SANS, Uniform RW-SANS are our reproductions using codes in [A		
M2ixKG.md	Uniform SANS	prose+table	and the results of ComplEx for Uniform, Uniform SANS, Uniform RW-SANS are our reproductions		
MDNCaching.md	MDNCaching	prose+table	for updates](page_1_image_1_v2.jpg) # MDNCaching: A Strategy to Generate Quality Negativ		
MDNCaching.md	SANS	prose+table	tly, Structure Aware Negative Sampling (SANS) [1] strategy was introduced with a dif		
NMiss.md	Frequency-based Negative Sampling	table	<table> <thead> <tr> <th>Frequency</th> <th>train</th> <th>		
PNS.md	Frequency-based Negative Sampling	table	elation Index</td> <td>Triplets Frequency</td> </tr> <tr> <td>0</		

Article	Entité	Où	Extrait		
RUGA.md	Relational Sampling	table	tr> <th>Entity</th> <th>Relational type</th> <th>Fact</th> </tr>		
SANS.md	RW-SANS	prose+table	combat this inefficiency, we introduce RW-SANS in Alg. 1, which uses ω random w		
SANS.md	SANS	prose+table	pose Structure Aware Negative Sampling (SANS), an inexpensive negative sampling stra		
SANS.md	Self-Adv	prose+table	to this technique as <i>Self-Adversarial (Self-Adv.) SANS</i> , whereas the former approach is		
SANS.md	Self-Adv. RW-SANS	prose+table	ted in Table 2 under Self-Adv. SANS and Self-Adv. RW-SANS, both of which reweigh the negative tri		
SANS.md	Self-Adv. SANS	table	ng></td> </tr> <tr> <td>Self-Adv. SANS (ours)</td> <td>52.03</		
SANS.md	Uniform SANS	prose+table	the former approach is referred to as <i>Uniform SANS</i> . Algorithm 1 Approximating the k		
SelfAdv.md	Bernoulli Negative Sampling	table	0</td> </tr> <tr> <td>TransH (bern)</td> <td>.668</td> <td>.87		
SNS.md	Random Negative Sampling	table	<td rowspan="4">TransH</td> <td>Random</td> <td>0.1520</td> <t		
STC.md	Frequency-based Negative Sampling	table	<thead> <tr> <th>Relation Frequency</th> <th>Test Number</th>		
TANS.md	Frequency-based Negative Sampling	table	<th>Index</th> <th>Query Frequency</th> <th>Answer Frequency</th>		
TANS.md	None Sampling	table	2">ComplEx</td> <td rowspan="3">None</td> <td>NS</td> <td>23		
TANS.md	SANS	prose+table	ike Self-Adversarial Negative Sampling (SANS) and subsampling. However, it is uncert		
TANS.md	Subsampling	table	tbody> <tr> <td rowspan="3">Subsampling</td> <td>Base</td> <td>		
TypeAugmented.md	Frequency-based Negative Sampling	table	Bin Center</td> <td>TaKE-Simple Frequency</td> <td>Simple Frequency</td>		
TypeConstraints.md	None Sampling	table	Freebase-150k</td> <td>None</td> <td>0.327</td> <td>		

C2. Candidats faux négatifs - MENTIONNÉS (30)

Article	Entité	Où	Extrait
CAKE.md	NS-KGE	prose	iciency of sampling. (5) None-sampling: NS-KGE (Li et al., 2021) eliminates the NS pro
CCS.md	IGAN	prose	sing an adversarial training framework. IGAN [40] and Kbgan [41] both introduce GAN

Article	Entité	Où	Extrait
ConceptDriven.md	CAKE	prose	Commonsense-Aware Knowledge Embedding (CAKE) framework, which consists of automatic
ConceptDriven.md	MVLP	prose	ule and the multi-view link prediction (MVLP) module, wherein CANS is comparable wit
dans.md	Self-Adv	prose	d other negative samplers (Random, Pop, Self-Adv and MCNS) employ RGCN [26] as the backb
DeMix.md	CAKE	prose	CANS [16]. The CANS is a component of CAKE [16] responsible for solving the invali
DeMix.md	MixGCF	prose	eneralization and robustness of models. MixGCF [13] integrates multiple negative sampl
DMNS.md	SANS	prose	g high-variance samples [9]. On graphs, SANS [2] select negative examples from k -h
DNS.md	Approximate Nearest Neighbor Negative Sampling	prose	an open source library that implements Approximate Nearest Neighbor functionality using Locality-Sensitive
DNS.md	Local-Closed World Assumption Negative Sampling	prose	ve samples, which are created under the Local Closed World Assumption (LCWA). Leveraging this assumption, one
ERDNS.md	Self-adversarial Negative Sampling	prose	the loss function in Self-Adv [21] and SANS [1]. However, these approaches primaril
eTruncatedUNS.md	Truncated Negative Sampling	prose	argin-based ranking loss. ϵ-Truncated Uniform Negative Sampling A widely-u
GibbsNS.md	IGAN	prose	distributions in the training process. IGAN [19] designed a generator and performed
GibbsNS.md	Nearest Neighbor Sampling	prose	e original feature space, such as the K-Nearest Neighbor (KNN) algorithm [30], which calculates
GNS.md	Local-Closed World Assumption Negative Sampling	prose	e inconsistent triples. In addition, a Local-Closed World Assumption (LCWA) approach [10] assumes a KG to be
GraphGAN.md	IRGAN	prose	he result suggests that, different with IRGAN (Wang et al. 2017a), the design of grap
HaSa.md	Structure-Aware Negative Sampling	prose	rxiv:2305.10563v2 # HaSa: Hardness and Structure-Aware Contrastive Knowledge Graph Embedding
KBGAN.md	IRGAN	prose	using policy gradient and other tricks. IRGAN (Wang et al., 2017) is a recent work wh

Article	Entité	Où	Extrait
M2ixKG.md	Entity Similarity-based Negative Sampling	prose	nition of hard negative samples Correct Entity Similarity based Hard Negative Samples (HNS-CES).
M2ixKG.md	MixGCF	prose	use mixing for harder negative mining. MixGCF [Huang et al., 2021] uses positive mixi
M2ixKG.md	Self-adversarial Negative Sampling	prose	-quality negative samples. Differently, SANS [Ahrabian et al., 2020] absorbs graph s
ReasonKGE.md	SANS	prose	ents structure-aware negative sampling (SANS), which utilizes the graph structure by
ReasonKGE.md	Local-Closed World Assumption Negative Sampling	prose	on the (local) closed world assumption (LCWA). Based on CWA all triples not present
SANS.md	Uniform RW-SANS	prose	heme. Figure 2: The performance of Uniform RW-SANS with TransE on FB15K-237 using differen
SNS.md	SANS	prose	hing, structure aware negative sampling(SANS) are four negative sampling methods in
SparseNSG.md	Frequency-based Negative Sampling	prose	# A Novel Negative Sampling Based on Frequency of Relational Association Entities for
STC.md	Local-Closed World Assumption Negative Sampling	prose	low the local closed-world assumptions (LCWA) proposed in [Krompaß et al., 2015] as
TuckerDNCaching.md	ESNS	prose	ity Similarity-based Negative Sampling (ESNS) (Yao et al., 2022) considers semantic
TuckerDNCaching.md	SANS	prose	ors, Structure Aware Negative Sampling (SANS) (Ahrabian et al., 2020) shows better p
TuckerDNCaching.md	Self-Adv	prose	., 2019) introduced a self-adversarial <i>Self-Adv</i> sampling approach based on a self-scor